

MADHUKA VIRAJITH

Software Engineering Undergraduate | Full-Stack & AI/ML Development

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SUMMARY

Software Engineering undergraduate at Cardiff Metropolitan University (BSc Hons in progress; Higher Diploma completed with Merit) with hands-on experience building eight full-stack, web, AI/ML, desktop, and mobile applications, applying knowledge to real performance problems. Seeking a Software Engineering internship to contribute to the industry.

TECHNICAL SKILLS

Java, Python, JavaScript, React, SQL, Data Structures & Algorithms, OOP, Git/GitHub

PROJECTS

Personal Projects

Tutor Finder - AI Powered Tutor Booking & Recommendation Platform

Feb 2026 - Jul 2026

Stack: Java, Spring Boot, Spring AI, Spring Security, React, Tailwind CSS, Google Gemini API

GitHub: <https://github.com/madhukavirajith/tutor-finder-frontend> | Live Demo: tutor-finder-frontend.vercel.app

- Implemented a custom Trie with Depth-First Search in Java for real-time, case-insensitive subject autocomplete.
- Designed a hybrid recommendation engine using a Max-Heap (PriorityQueue) to rank the top K tutors in $O(K \log N)$ time, with Levenshtein Distance for fuzzy search.
- Integrated a context-aware AI chatbot (Spring AI + Google Gemini) with topic guardrails; secured the platform with JWT/Spring Security role-based access for students, tutors, and admins.

LeaveTrackerPro - Desktop Employee Leave Management System

Jan 2026 - Jun 2026

Stack: C#, .NET 8, Windows Forms, SQLite

GitHub: github.com/madhukavirajith/LeaveTrackerPro

- Built an enterprise-style desktop app supporting leave requests, manager approvals, and admin policy management with immutable audit logs.
- Implemented leave-balance tracking across 6 leave types, a team-availability calendar, Excel export for HR reporting, and BCrypt password hashing.

Academic Projects

MEDA Mars Atmospheric Pressure Predictor - Virtual Sensor Recovery for NASA Perseverance

Jun 2026 - Aug 2026

Stack: Python, Streamlit, XGBoost, scikit-learn, Pandas, NumPy, Matplotlib

GitHub: github.com/madhukavirajith/mars-pressure-predictor | Live Demo: mars-pressure-predictor.streamlit.app

- Built an ML web app that reconstructs Martian atmospheric pressure from 23 correlated MEDA telemetry channels (temperature, humidity, solar irradiance, rover kinematics) from NASA's Perseverance Rover at Jezero Crater.
- Trained and serialised an XGBoost regression model with a scikit-learn imputation pipeline to handle missing sensor data in real time, enabling virtual recovery of a failed pressure transducer.
- Built an interactive Streamlit dashboard with preset mission scenarios and a zoned pressure gauge; deployed on Streamlit Cloud with cached model loading for low-latency inference.

CarbonWise SL - AI Powered Household Carbon Prediction & Reduction Platform

Apr 2026 - Present

Stack: React.js, FastAPI, Python, Docker, Firebase, XGBoost, SHAP

GitHub: github.com/madhukavirajith/CarbonWiseSL | Live Demo: carbon-wise-sl.vercel.app

- Built an XGBoost regression model to predict household daily carbon emissions, with SHAP TreeExplainer for per-appliance explainability.
- Applied K-Means clustering to segment consumption patterns and generate personalised energy-reduction recommendations.
- Developed a Solar ROI calculator forecasting installation cost, payback period, and lifetime carbon offset from regional irradiance data.

Sunrise Dental Clinic - Appointment & Patient Management System

Jul 2025 - Sep 2025

Stack: Java 17, Jakarta EE (Servlets, JSP, JSTL), MySQL, Maven, JUnit 5, GitHub Actions

GitHub: github.com/madhukavirajith/sunrise-dental-clinic-system

- Engineered a 3-tier Java EE web application with no third-party frameworks, applying DAO, Singleton, Strategy, and Observer patterns to separate presentation, business logic, and data access.
- Used Strategy for treatment-dependent billing and Observer for a simulated Email/SMS notification system with a full audit trail.
- Secured the app with PBKDF2-salted password hashing and a servlet-filter session check, and exposed a hand-built REST-style JSON endpoint.
- Implemented stored procedures, functions, and triggers (e.g. double-booking prevention); wrote 65 unit, integration, and boundary-value tests with CI via GitHub Actions.

Forgotten Recipes - MERN-Stack Heritage Culinary Platform

Jun 2025 - Aug 2025

Stack: MongoDB, Express.js, React, Node.js, Socket.IO

GitHub: github.com/madhukavirajith/Forgotten-Recipes | Live Demo: forgotten-recipes.vercel.app

- Led a team of 6 as Product Owner and Scrum Master, running Agile sprint planning and backlog management to deliver a full-stack MERN application.
- Built real-time chat between home cooks and chefs with Socket.IO, and a nutrition visualiser with macro/micro-nutrient radar charts.
- Secured the platform with JWT/Bcrypt authentication, Helmet security headers, and Express rate limiting.

FitZone Fitness Center Portal - Booking & Management System

Feb 2025 - Apr 2025

Stack: PHP, MySQL, HTML5, CSS3, JavaScript, Docker

GitHub: github.com/madhukavirajith/fitzone | Live Demo: fitzone-0sp0.onrender.com

- Built a secure PHP backend with session-based authentication and parameterised SQL queries to prevent SQL injection, backed by a relational MySQL schema.
- Designed separate customer and staff/admin dashboards for bookings, inquiries, and appointment logs.
- Containerised the stack with Docker Compose and deployed it on Render with a cloud-hosted database on Clever Cloud.

Luxe Vista Resort - Android Hotel Management System

Feb 2025 - Apr 2025

Stack: Java, Android SDK, SQLite

GitHub: github.com/madhukavirajith/LuxeVistaResort

- Architected the app with MVVM and applied the Singleton pattern to manage a single shared database connection.
- Used RecyclerView with background data processing to keep large room/service lists responsive, and Glide for efficient image loading.
- Built local storage for offline-reliable user profiles and reservation history.

PROFESSIONAL EXPERIENCE

Data Annotator - Innodata Inc.

Jan 2026 - Jun 2026

- Labelled data to strict quality guidelines for a Superintelligence Lab annotation project supporting large-scale AI model training, maintaining high accuracy under time-sensitive targets.

Data Annotator - IFG BPO (Pvt.) Ltd.

Oct 2025 - Mar 2026

- Processed and analysed urban data for the Greehill project, collaborating with a remote team to meet deadlines with consistent quality.

EDUCATION

BSc (Hons) Computer Software Engineering - Cardiff Metropolitan University

Nov 2025 - Present

- Key modules: Advanced Programming, Analytics and Business Intelligence, Computational Intelligence, Professional and Ethical Issues in IT, Development Project

Higher Diploma, Computing & Software Engineering - Cardiff Metropolitan University

Jan 2024 - Nov 2025

- Grade: Merit. Key modules: Data Structures & Algorithms, OOP, Database Design & Development, Web & Mobile Application Development, Service-Oriented Computing, System Analysis and Design, Computer Networks, Project Management

CERTIFICATIONS

- Google AI Essentials Specialization - Google, Dec 2025 (Credential ID: CRVK7DM1HRYO)
- Java Programming for Beginners - IBM, Sep 2025 (Credential ID: QKQ28D53HMVA)